

Senior MLOps / Machine Learning Engineer with 7+ years building and operating production ML systems end to end — data, models, deployment, and monitoring. Currently own 12+ ML and GenAI microservices on Azure Kubernetes processing 20,000+ images and documents daily for real-time insurance-claims automation: computer vision pipelines, image fraud detection, LLM document intelligence, and agentic AI platforms. Strong engineering fundamentals — async Python, SQL and NoSQL data systems, streaming and batch processing, Docker, CI/CD — with a record of trade-off-driven design balancing latency vs. accuracy and cost vs. performance at scale.

SKILLS

Languages & Databases	Python (async/concurrency), SQL, NoSQL (Azure Cosmos DB, MongoDB, DynamoDB), C++, Shell Scripting, Pandas, NumPy
ML & GenAI	PyTorch, TensorFlow, YOLOv8, Mask2Former, CLIP embeddings, OpenCV, scikit-learn, LangChain, LangGraph, RAG, MCP (FastMCP), Azure AI Foundry, OpenAI GPT-4o (vision), Azure Document Intelligence, model monitoring & drift detection
Streaming & Batch	Azure Service Bus queue workers, Batched GPU inference, bounded-concurrency pipelines (asyncio/aiohttp), event-driven architectures (AWS EventBridge, Step Functions)
Infrastructure & DevOps	Docker (GPU/CUDA), Kubernetes (AKS), Helm, CI/CD (Azure Pipelines, GitHub Actions, CodePipeline), Terraform, Ansible, Azure & AWS, Nginx, Application Insights observability, Linux, Git
Communication	English, Urdu

TECHNICAL EXPERIENCE

Senior MLOps Engineer

Trilateral Research

October 2022 — Present

Remote — UK

- Solely designed and built the backend of a customer-facing no-code AI-agent platform (FastAPI, Langgraph, Cosmos DB) — MCP tool federation, RAG knowledge bases, voice agents, human-in-the-loop approvals, per-tenant token budgets — onboarding 10 tenants with 36 published agents within two weeks of launch; also shipped a production MCP server enabling agent-driven service bookings.
- Productionized a LangGraph text-to-SQL agent over four operational SQL Server databases (parallel querying, autonomous error-repair resolving >90% of failed generations), delivering 300+ self-service reports/day to non-technical staff.
- Build and operate 12+ production ML/GenAI microservices (Python, FastAPI, Docker, AKS) automating the vehicle-insurance claims lifecycle end to end — owning data flows, model integration, deployment, and monitoring for systems processing 20,000+ images and documents daily.
- Single-handedly built a multi-tier image fraud detection service (PyTorch, CLIP embeddings, perceptual hashing) screening 7,000–12,000 claim images/day against a ~1M-image embedding store in Cosmos DB at ~90% fraud-catch rate and ~5% false positives; tuned similarity thresholds on production cosine-score distributions to balance missed fraud against false alarms.
- Architected the entire production system (all but model training) for a 6-microservice computer-vision damage-analysis platform serving 5,000–8,000 inspections/day; designed the async orchestration layer fanning inference out in parallel, making the pipeline ~5× faster than sequential execution.
- Owned end-to-end an async vehicle-information extraction service (YOLOv8 + GPT-4o vision OCR) processing up to 10,000 images/day at <5 s latency, raising extraction accuracy from ~80% to ~95% over the legacy OCR-only system.
- Architected a document-intelligence product (Azure Document Intelligence + GPT-4 vision) processing 5,000+ accident reports/month within a 24-hour SLA; drove the vision-LLM-over-OCR design decision through an accuracy/cost trade-off analysis (~20% accuracy gain, ~200 client review hours/month saved).
- Built queue-driven streaming and batch processing pipelines (Azure Service Bus workers, batched GPU inference, asyncio + ProcessPoolExecutor hybrid concurrency) decoupling ingestion from compute — e.g. video-inspection processing in 1–3 min/video with ~30% lower inference cost via frame-sampling and batch tuning.
- Solely built and maintain the team's model-monitoring platform (Streamlit, Cosmos DB) tracking ~16 production CV models across versions for regressions and drift — caught a pre-release F1 regression enabling rollback before production impact and cut regression-discovery time by ~50%.
- Standardized all services on templated Azure Pipelines CI/CD to AKS with GPU/CUDA containers, structured logging, and cross-service distributed tracing via Application Insights custom dimensions.
- Architected end-to-end ML pipelines across all products, mentored junior engineers, and partnered with data-science teams on experimentation and deployment while continuously optimizing AWS infrastructure costs.
- Overhauled legacy Airflow ML pipelines into an event-driven architecture (AWS EventBridge, Step Functions, Lambda, ECS Fargate, GPU EC2), cutting training-to-deployment from 7 days to 2 and enabling multiple daily releases vs. 1–2 per month.
- Engineered parallel training and batch inference for 6 models across 3 product lines processing up to 50K records daily, reducing infrastructure costs by 30%.

- Developed an ML monitoring system tracking data drift, concept drift, feature attribution, and bias for 6 production models, with stakeholder dashboards enabling proactive retraining and preventing multiple model failures.
- Designed a YAML-based dynamic CI/CD framework (AWS CodePipeline, GitHub Actions) adopted by 3 teams (35+ developers), cutting build times from 3.5–4 hours to 5–8 minutes; built reusable Terraform modules reducing new-environment setup from 2–3 weeks to 2–3 days.
- Introduced build-once-deploy-anywhere, one-click rollbacks, and branch-agnostic deployments, accelerating development cycles and developer productivity.
- Created reusable Terraform high-level modules for core AWS services (Lambda, ECS, API Gateway, RDS, etc.), hosted in GitHub, and established environment blueprints that reduced new account infrastructure setup from 2–3 weeks to 2–3 days across 12 downstream stacks.
- Refactored monolithic state files into modular job-based stacks, streamlining updates and maintenance.

ML Engineer

March 2019 — September 2022

Urbanise

Melbourne, Australia

- Led the Machine Learning team delivering end-to-end ML and big-data solutions with automated CI/CD and deployment pipelines on AWS (SageMaker, ECS, Lambda, API Gateway, Terraform).
- Developed an AI-based duplicate work-order detection system (TensorFlow/TF Serving) with a fully automated AWS deployment pipeline, reducing workforce costs by automatically flagging duplicate jobs.
- Shipped real-time text completion for complaint descriptions and an image-captioning system generating descriptions from photos, with end-to-end preprocessing, training, and inference pipelines provisioned via Terraform.
- Built a conversational chatbot (RASA + Amazon Polly) enabling customers to raise work orders and check ticket status through natural language, reducing support-team workload.
- Developed custom image captioning system enabling clients to generate complaint descriptions directly from photos of issues, streamlining reporting workflows. Integrated computer vision models with backend services using serverless architecture on AWS.
- Designed Deep learning based template and supplier assignment system for facilities management jobs, enhancing manual human-in-the-loop processes and improving assignment accuracy. Leveraged TensorFlow and AWS services for scalable, automated decision-making.

EDUCATION

Masters in Data Science, *National University of Computer & Emerging Sciences, Islamabad*

July 2023

Bachelor of Science in Electronics Engineering, *COMSATS University Islamabad*

June 2019

Publication, BlazePose-Based Action Recognition with Feature Selection Using Stochastic Fractal Search

2022 — 2023

Master Thesis, Spatio-Temporal Scene Graph Generation

2022 — 2023

Bachelor's Final Year Project, Urdu Language Chatbot Using Seq2Seq with Bahdanau Attention Mechanism

2018 — 2019